

Robot Framework — Task 2

1. Experiment Title

Variables and Data-Driven Automation Using Robot Framework

2. Problem Statement

Implement a Robot Framework test case that uses variables to store login data such as usernames and passwords, and create a data-driven test that reads test data from an external CSV file and performs the same login actions on multiple sets of data.

3. Objective

To demonstrate the use of **Robot Framework variables** and **DataDriver** for executing the same test case with multiple sets of external test data.

4. Tools, Software and Concepts Used

Tools / Software

- Python
- Robot Framework
- SeleniumLibrary
- Robot Framework DataDriver
- Google Chrome
- Visual Studio Code
- CSV file

Concepts

- Robot Framework variables
- Data-driven testing
- External CSV test data
- DataDriver library
- Test Template
- Selenium WebDriver
- Test Teardown
- Conditional keyword execution
- Login validation

5. Source Code / Implementation Steps

File 1: login_test.robot

*** Settings ***

Library SeleniumLibrary

Library DataDriver file=login_data.csv encoding=utf-8

Test Template Login Test

Test Teardown Close All Browsers

*** Variables ***

\${URL} https://www.saucedemo.com/

*** Test Cases ***

Login Test With Different Data

 \${username} \${password} \${expected}

*** Keywords ***

Login Test

 [Arguments] \${username} \${password} \${expected}

 Open Browser \${URL} Chrome

 Maximize Browser Window

 Input Text id=user-name \${username}

 Input Text id=password \${password}

 Click Button id=login-button

 Run Keyword If '\${expected}' == 'success'

 ... Location Should Contain inventory.html

 Run Keyword If '\${expected}' == 'error'

 ... Page Should Contain Element css=h3[data-test="error"]

 Log Username: \${username}

 Log Expected Result: \${expected}

File 2: login_data.csv

The final working CSV uses semicolon separators:

```
*** Test Cases ***;${username};${password};${expected}
```

```
Valid Login;standard_user;secret_sauce;success
```

```
Invalid Username;wrong_user;wrong_password;error
```

```
Invalid Password;standard_user;wrong_password;error
```

Implementation Steps

1. Import SeleniumLibrary and DataDriver.
2. Define \${URL} as a Robot Framework variable.
3. Configure DataDriver to read login_data.csv.
4. Use Test Template to execute the same Login Test keyword for each data row.
5. Store username, password, and expected result in variables.
6. Open SauceDemo and enter the supplied credentials.
7. For successful login, verify inventory.html.
8. For invalid login, verify the error element.
9. Use Test Teardown to close all browsers after each test.

6. Output with Clearly Labelled Screenshots

